

Integration of On-skin Interface into Cave Automatic Virtual Environment (CAVE-like System)

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ABSTRACT

This paper proposes to integrate an emerging technology, on-skin interface, into the Cave Automatic Virtual Environment (CAVE-like system) to enhance immersion experience and contribute to grasping rehabilitation. The CAVE-like system can create an engaging rehabilitation environment which will prevent boredom and monotony and make the process more attractive. Nowadays, the prevailing method of interacting with the CAVE-like system relies on traditional handheld controllers but reveals limitations. Holding controllers detaches users from the virtual world as the controllers create a physical barrier between users and the virtual environment. Moreover, handheld controllers are unsuitable for individuals with special needs requiring grasping rehabilitation because they have difficulties manipulating handheld devices. The on-skin interface is expected to hold great promise in enabling hands-free interaction, thereby achieving a more natural human-computer interaction. This paper presents a project plan for developing and implementing an on-skin interface within a CAVE-like system to enhance grasping rehabilitation, while also considering other potential applications.

Index Terms: Hardware—Emerging technologies—Emerging interfaces; Human-centered computing—Human computer interaction (HCI)—Interactive techniques—Gestural input; Hardware—Communication hardware—interfaces and storage—Wireless devices

1 INTRODUCTION

The Cave Automatic Virtual Environment (CAVE-like system), a projection-based Virtual Reality display [1] as shown in Figure 1, provides an immersive environment for users to experience the virtual world. With the assistance of the tracking system, the images in the CAVE-like system can be dynamically adjusted based on the direction and position of the tracker worn by the viewers, enabling realistic movement and navigation within the virtual environment. The CAVE-like system has experienced significant development and widespread application in various domains, such as gaming and training. Notably, the CAVE-based game "RehabMarket" [2] has demonstrated promising outcomes in supporting post-stroke rehabilitation programs according to the positive user feedback. Additionally, C. Tuena et al. [3] has also showcased the efficacy of CAVE-like systems for spatial navigation rehabilitation. It could be seen that the utilization of CAVE-like systems expands in rehabilitation programs and there is an application of grasping rehabilitation under development.

During the development of content for grasping rehabilitation, the fact that the handheld controllers may not be applicable for the

targeted patients bought content developers' attention. Developers have made efforts to find alternative methods to replace handheld controllers, such as utilizing real models with trackers to represent the target objects. However, the real models were revealed as an infeasible solution because the trackers may become obstructed or covered during user interactions, affecting normal operation of the tracking system. Consequently, using real models cannot be a solution.



Figure 1: The CAVE-like system located in the lab of the department of industrial and manufacturing system engineering (IMSE lab) at the University of Hong Kong.

To address the difficulty, the emerging technology of on-skin interface attracts developers' attention and are regarded as a promising solution. By integrating the device into the CAVE-like system, users can engage in an immersive grasping experience without the constraints imposed by traditional handheld controllers or tracking systems. The on-skin interface device represents an innovative technology that enables users to interact with virtual environments through touch and gesture commands. There is an on-skin interface developed by Tianhong Catherine Yu et al. [5] composed with a self-powered sensor and a micro-controller as shown in Figure 2 to harvest mechanical energy, convert mechanical energy into electrical signals and deliver signals wireless to computer system.

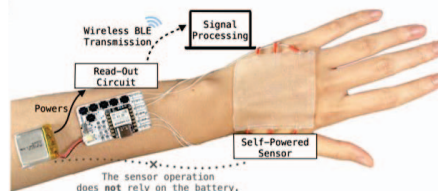


Figure 2: Overview of on-skin interface device developed by Tianhong Catherine Yu et al. [5].

Generally, the composition of on-skin interface developed in the paper is similar to the composition invented by Tianhong Catherine Yu et al. [5]. The microcontroller plays a critical role in receiving, analyzing, and interpreting electric signals generated by the self-powered sensors, and then delivers command issuance to the

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computer. The core functional parts of the self-powered sensors are the nanogenerators, which could operate without an electric source. By harnessing the principles of triboelectric effects, nanogenerators can efficiently convert forces into electric signals. Consequently, the device becomes responsive to user actions through generating electric signals when users touch the self-powered sensor. The self-powered sensor is manufactured by machine-embroidery which is regarded as a low-cost, convenient, and flexible design method [5]. However, with the development of 3D printing technology, 3D printing is proposed as another low-cost, convenient, and flexible manufacturing method because 3D printing allows the device to be printed by different materials as needed and personalized shape.

The paper presents a project plan to demonstrate the functionality of the on-skin interface within the XRCC content creator [4] developed by Votanic Company, utilizing the CAVE-like system. The selected platform, widely used by the lab of the department of industrial and manufacturing System Engineering (IMSE lab) for training projects, offers a user-friendly software solution for individuals without programming knowledge. The primary objective is to integrate the on-skin interface seamlessly into the CAVE-like system, initially focusing on implementing a basic grasp function. By leveraging the versatile structures of the device, which generate diverse electrical signals, and exploring the potential of combining the device with more unexplored content, the project would like to open a wide range of possibilities. The envisioned future applications of on-skin interface devices hold promise for enhancing interactions in virtual environments and venturing into new realms, presenting numerous exciting opportunities for exploration and innovation.

2 DESIGN AND IMPLEMENTATION

This paper will present the prototype design and provide an overview of the steps that have been completed or are currently underway.

2.1 On-Skin Interface Design

Figure 3 illustrates the design overview of the on-skin interface. The self-powered sensor is strategically placed at the middle fingertip. When a grasping action occurs, the middle fingertip and thumb fingertip pinch together, exerting pressure on the sensor. The pressure will be detected by the sensor, which generates corresponding electrical signals, which will then be transmitted to a microcontroller for interpretation. Based on the interpreted signals, the microcontroller issues instructions to the CAVE-like system, enabling seamless integration between the on-skin interface and the virtual environment.

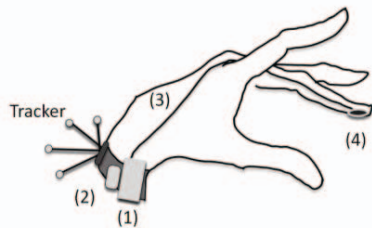


Figure 3: Design overview of the on-skin interface prototype with one tap. The prototype is composed of (1) a microcontroller, (2) a battery, (3) a wire and (4) a self-powered sensor. The stuff on the wrist is a tracker shown in Figure 3, which assists the tracking system locate the position of the hand.

2.2 Implementation

The project is currently ongoing, and a significant milestone has been reached with the development of a 3D printed Polytetrafluoroethylene (PTFE) nanogenerator. Jiaming Zhou, a PhD student from the University of Hong Kong's Department of Mechanical Engineering, has played a crucial role in supporting this development. To evaluate the functionality of the nanogenerator, testing was conducted using an electric light as a means of verification. The testing aimed to assess the nanogenerator's ability to generate electricity through mechanical deformation. The successful testing with the electric light demonstrated the potential and effectiveness of the nanogenerator in generating electrical energy.

2.2.1 Polytetrafluoroethylene (PTFE) Nanogenerator

The project commenced with a nanogenerator 3D printed with polytetrafluoroethylene (PTFE) material. PTFE has been discovered to its potential as an efficient energy harvester by successfully converting mechanical energy into electric signals with high output [6] and PTFE has been proven to be a practical 3D printing material [7, 8]. The choice of material ensures clear and conclusive results at the beginning stage, allowing for apparent evaluation of the nanogenerator's performance and capabilities.

2.2.2 Testing PTFE Nanogenerator

Before applying the on-skin interface to the XRCC, it is prerequisite to evaluate the functionality of the 3D printed PTFE prototype. A light testing method and a voltmeter were employed. One end of the light was connected to the PTFE nanogenerator, while the other end was connected to the ground side, which can be safely held by hand to prevent short circuits. During the test, when a user tapped the PTFE nanogenerator with a finger, the light connected to the setup instantaneously illuminated, indicating the successful generation and transmission of electrical signals by the nanogenerator.

Then, a voltmeter was applied to measure the peak voltage and stability of the voltage to ensure the microcontroller can capture the electric signals properly. The testing is still in progress. At the present stage, the plan is to apply an operational amplifier to increase the generated voltage signals. The collected voltage data was not enough to make a conclusion and more systematic testing will be carried out. The tests will serve as practical demonstrations of the prototype's capability to convert mechanical energy into electrical energy, validating the functionality.

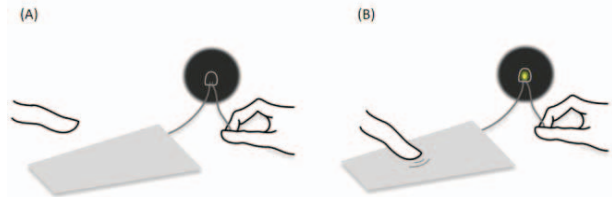


Figure 4: (A) PTFE nanogenerator with a small electric light. (B) When a user taps the PTFE nanogenerator with a finger, a small electric light connected to the setup will instantly illuminate.

3 FURTHER WORK

The ongoing testing serves as evidence for the feasibility of the self-powered sensor. While progress is promising, there are still multiple steps remaining to fully complete the prototype.

3.1 Scenario Development in XRCC

The subsequent step involves developing a scenario using the XRCC content creator. The scenario is planned to place the user in front of a desk within the virtual environment, which is a basic environment setup. Given that many rehabilitation patients, particularly elderly individuals, face limitations in their ability to travel, a CAVE-like system can serve as a valuable tool to help them explore the world virtually. The CAVE-like system offers patients the opportunity to engage in virtual tourism experiences, enabling them to "visit" and explore various locations from the comfort of their own surroundings. Incorporating a simple storyline within the virtual environment can also enhance the grasping experience for patients during their rehabilitation.

3.2 Build PTFE Prototype

The PTFE nanogenerator will be connected to a microcontroller. Therefore, the signals generated by the PTFE nanogenerator can be captured and analyzed by a microcontroller.

3.3 Testing in the CAVE-like System

The final step involves testing the integrated prototype in the CAVE-like system to evaluate the performance and effectiveness of the on-skin interface within a fully immersive, projection-based VR system. In the system's backstage, specific commands will be set to interpret and respond to the generated signals effectively to establish seamless communication. The users need to wear a tracker as shown in Figure 4 on the wrist to locate the position of hand and tap the sensor to give instructions. The functionality will allow users to simulate the grasping actions to manipulate virtual objects by tapping on the on-skin interface and the movement captured by tracking system. The goal is to ensure a smooth and intuitive user experience, where users can seamlessly engage with the virtual environment, enhancing immersion and interactivity. After confirming the feasibility of the on-skin interface through testing with the ordinary people, it would be necessary to invite patients in need of grasping rehabilitation to participate in the project. The involvement of professional staff, such as rehabilitation therapists or clinicians, is crucial during this phase. Their expertise and guidance are essential for providing accurate assessments and suggestions for the final analysis of the project.



Figure 5: The tracker to locate the position in CAVE-like system in the lab of the department of industrial and manufacturing system engineering at the University of Hong Kong.

3.4 Improvement

After confirming the feasibility and functionality of the PTFE prototype, the project will focus on optimization of the prototype. Further improvements for on-skin interfaces can concentrate on the advancements in both sensor technology and content development. Enhancements in sensor materials and designs can improve accuracy and durability, while refining the integration of the interface with virtual environments or applications can create more immersive experiences.

To optimize the efficiency of the self-powered sensor, it is inevitable to conduct additional investigation and testing on the materials regarding the characteristics and the influential factors. With advancements in nanogenerator materials, there is a growing pool of options that exhibit positive results during function ability testing [9, 10, 11]. For example, Thermoplastic Polyurethane (TPU) material is considered prospective for its flexible and stretchable properties compared with PTFE material, and the enhanced tribo-positivity to further optimize the energy harvesting capabilities [9]. Therefore, embedding on-skin interface in textiles can be possible if it is necessary. Instead of a plate of hard material, the flexible and stretchable material could bring more chances to the wearable devices. It is also important to confirm and assess the fatigue life, known as durability, of the materials to ensure the long-term functionality. Additionally, the factors that may affect the operation of the on-skin interface need to be further studied. For example, the situation that dielectric constant appears clearly differences between male and female and the skin type has minor influences in conductivity value [12] may influence the generated electric signals. More and more influencing factors will be realized during the practical application process.

What smore, designing different shapes of the self-powered sensor will lead to diverse electric signals when sliding across the sensors. The diverse electric signals provide opportunities to generate more different and personalized commands to match personalized requirements to computer systems. As a result, how to design the user interface mechanism to take advantage of the signals are important for users' immersion experience.

4 CHALLENGE AND FUTURE DIRECTIONS

Because the on-skin interface is an under developing concept, everything is the first attempt in the CAVE-like system and there must be numerous potential challenges and opportunities as well. The paper will share several challenges that have encountered or anticipate encountering during the developing process. About the future application scenarios for the interface include but not limited to the CAVE-like system, several anticipatory visions are shared.

4.1 Challenge

In the technical domain, there are several aspects that require further exploration. For the generated electric signals, there are several concerns. The first challenge is to ensure accurate sensing. The electric signal generated is an instant signal as shown in Figure 6 which require the microcontroller to capture the signals within several microseconds correctly and precisely. The microcontroller must be guaranteed to not only capture the instant signals generated by the sensor but also capture the signals accurately, especially considering the potential increase of diversity of electric signals led by the complexity of the sensors in the future. Secondly, the stability and performance of the sensors to generate electric signals over long periods of time and under different conditions should be further assessed.

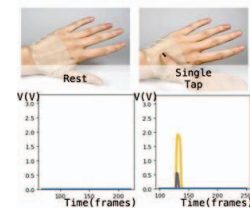


Figure 6: Testing results of generated instant electric signals for one tap by Tianhong Catherine Yu et al. [5].

In the realm of content development, whether the integration of on-skin interface in the CAVE-like system could enhance

immersion needs to be further investigated. The next step involves engaging numerous experiencers to gather feedback and insights. Especially for grasping rehabilitation, professional assessment and suggestions from rehabilitation center are significant to get a definitive conclusion and make adjustment. Specialized knowledge and guidance will ensure the on-skin interface and the content are well combined and helpful for rehabilitation.

4.2 Future Directions

Although it is not yet the moment when the on-skin interface could be popularly accommodated, the potential application scenarios for on-skin interface could be foreseen, which will motivate people to make efforts to make improvements in either virtual reality or application in daily life. Several example applications, such as music player patch, blink detection, remote control, and emergency notification stickers, shared by Tianhong Catherine Yu et al. [5], trigger developers' think about the more application scenarios of on-skin interface.

After integrating the on-skin interface into XRCC in CAVE-like system, other than grasping rehabilitation, other training or gaming applications could be further explored. The interface allows to add interactivity through physical gestures, touch, and haptic feedback. Exploring these applications unleashes the full potential of the on-skin interface in different scenarios, pushing the boundaries of training and gaming experiences. Moreover, it is worth testing the device in the Unity and Unreal engine, which are widely applied game engines that supports the development of majority games. If the on-skin interface is expected to be widely used, it is vital to figure out how it works in the popular game engines.

The on-skin interface could be integrated into other platforms such as VR headset and AR glasses. The possible applications could be sketching and remote control. For example, when wearing a VR headset, the way to input the text is to use wand led by controller to choose the letter form the virtual keyboard which is very inconvenient. If the on-skin interface allow to sketch and write on the skin of users' hand, it will be easy to give instructions from real body to the virtual world. Regarding the AR glasses, the common methods to give commands are pressing buttons or vocal instructions which limit the applications that could be used. The assistance of the on-skin interface should allow more practical applications.

5 CONCLUSION

Though the on-skin interface is still under development, the testing results till now are positive. The auxiliary of on-skin interface in the CAVE-like system is predicted to be beneficial for grasping rehabilitation and enhancing users' immersion experience.

The CAVE-like system will offer a more engaging and appealing rehabilitation environment, helping to prevent boredom and monotony associated with repetitive exercises. By immersing patients in a virtual world built through XRCC filled with diverse cultural settings and engaging activities, the rehabilitation process is supposed to become more attractive and interesting. And by introducing narrative elements and interactive scenarios, the grasping exercises should become more engaging and enjoyable. The approach may be helpful to alleviate the potential tedium and monotony of repeated grip exercises in real life, making the rehabilitation process more motivating and enjoyable for patients. Therefore, the patients are more likely to maintain interests and motivation, potentially leading to improved outcomes in their rehabilitation progress.

The on-skin interface will allow patients to simulate the grasping process. One of the advantages of the on-skin interface is that patients do not need to physically grab a traditional controller to provide commands to the system. Instead, patients can simply pinch fingertips of thumb and middle finger together and apply

pressure to the self-powered sensor to generate electrical signals, which means they successfully achieve a grasp action. The signals detected and analyzed by a microcontroller will be transmitted to the system, which registers the command and translates it into corresponding movements of the virtual object within the virtual world. The hand tracker assists in accurately tracking the user's hand movements, ensuring that the virtual object moves accordingly.

The participation of professional staff is also significant because the professional staff can offer valuable insights into the effectiveness of the on-skin interface in the context of grasping rehabilitation. They can assess the impact of the interface on patients' motor skills, overall engagement, and progress. Their feedback and recommendations can help fine-tune the project and make any necessary adjustments to ensure that the on-skin interface meets the specific needs of the patients and aligns with established rehabilitation protocols.

The integration of on-skin interface into CAVE-like system is anticipated to contribute to the grasping rehabilitation for people in need. During the process of exploring the novel technology, there must be lots of technical barriers including but not limited to ensure accurate sensing, designing flexible materials, optimizing power efficiency, and establishing seamless communication. User experience challenges, such as ergonomics and usability, are also important and challenging to be addressed. The on-skin interface holds immense potential and warrants further exploration in various directions. Ongoing research can focus on improving sensor accuracy, enhancing system integration, and exploring diverse applications beyond rehabilitation. Continued exploration and experimentation will unlock the full capabilities of the on-skin interface and lead to innovative advancements in multiple fields.

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